

# XIANG LI

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## EDUCATION

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| <b>Peking University, Beijing, China</b><br>Ph.D. in Statistics, School of Mathematical Sciences<br>Advisor: <a href="#">Prof. Zhihua Zhang</a>      | Sep. 2018 – Jul. 2023 |
| <b>Peking University, Beijing, China</b><br>B.S. in Statistics, School of Mathematical Sciences<br>B.A. in Economics, National School of Development | Sep. 2014 – Jul. 2018 |

## EXPERIENCE

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| <b>Rutgers University, Piscataway, USA</b><br>Assistant Professor, Department of Statistics                                                                     | Starting Jan. 2027    |
| <b>University of Pennsylvania, Philadelphia, USA</b><br>Postdoctoral Researcher. Working with <a href="#">Prof. Weijie Su</a> and <a href="#">Prof. Qi Long</a> | Aug. 2023 – Present   |
| <b>University of Toronto, Toronto, Canada</b><br>Visiting Ph.D. Student. Working with <a href="#">Prof. Qiang Sun</a>                                           | Feb. 2022 – Feb. 2023 |
| <b>Face++ (Megvii), Beijing, China</b><br>Research Intern, Algorithm Group. Mentor: <a href="#">Dr. Shuchang Zhou</a>                                           | Jun. 2018 – Nov. 2018 |
| <b>ToSimple, Beijing, China</b><br>Research Intern, Algorithm Group. Mentor: Dr. Naiyan Wang                                                                    | Mar. 2018 – May. 2018 |

## SELECTED AWARDS

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|------------------------------------------------------------------------------|-------------------------|
| • IMS New Researcher Travel Award                                            | 2025                    |
| • Outstanding PhD Graduates, Peking University                               | 2023                    |
| • President Scholarship for Excellent Ph.D Student, Peking University        | 2018, 2020-2021         |
| • Outstanding reviewer in NeurIPS, top 8%                                    | 2021                    |
| • National Scholarship, China                                                | 2017, 2019              |
| • Travel Award                                                               | NeurIPS 2019, AAAI 2020 |
| • Outstanding Graduates, School of Mathematical Sciences, Peking University  | 2018                    |
| • First prize in National Undergraduate Mathematical Modeling Contest, China | 2016                    |

## RESEARCH INTERESTS

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My research lies at the intersection of statistics, machine learning, and optimization. I develop statistical methodology and theory for modern data-scientific problems, with emphasis on inference, uncertainty quantification, robustness, privacy, and scalable learning algorithms. A major part of my recent work studies generative AI and large language models from a statistical perspective, including provenance, evaluation, uncertainty, and reliability. More broadly, my research spans the following directions:

- **Statistical Foundations of Modern Machine Learning**

- **Generative AI, Provenance, and Reliability.** Developing statistical methods for analyzing and verifying outputs from generative models, including watermarking schemes [J4, C3, J5, C4], detection procedures [J1, J2, C5, S3], and estimation methods [J3, S2] for provenance, accountability, and reliability.
- **Evaluation and Uncertainty Quantification.** Modeling complex learning systems as statistical black boxes and developing tools to study input–output behavior, with applications to model evaluation [S4, C9], uncertainty quantification [C8], scaling laws, privacy-preserving analysis, and distributional shift.
- **Reliable Statistical Machine Learning**
  - **Learning under Challenging Conditions.** Developing efficient algorithms for nonstandard settings such as heavy-tailed noise [J10], corruption [C10], high-dimensionality, and data heterogeneity, and analyzing their statistical and convergence properties [C17, C12].
  - **Inference in Adaptive and Dynamic Environments.** Designing statistical methods for streaming data [J6, J8] and adaptive decision-making [C11, C16, C18], with the goal of ensuring valid inference in dynamic, dependent, or data-adaptive environments [C2, T1, J7, S6].
- **Federated Learning, Privacy, and Distributed Statistics**
  - **Communication-Efficient Learning.** Developing communication-efficient algorithms [J11, C15, T3] and analyzing their convergence properties under heterogeneous data settings [C1, C13, T2].
  - **Trustworthy Distributed Learning.** Studying statistical and algorithmic aspects of trustworthy federated learning, including differential privacy [C6, S5], personalization [J9, C14], and robust distributed inference.

## PUBLICATIONS

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\* indicates equal contribution. \*\* indicates alphabetical order.

### Journal articles (Published or under revision)

- [J1] **A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules** [pdf] [arxiv]  
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su  
*Annals of Statistics*, 53 (1): 322–351, Feb. 2025  
Selected for presentation at the Annals of Statistics Invited Paper Session, JSM 2025.
- [J2] **Robust Detection of Watermarks in Large Language Models Under Human Edits** [pdf] [arxiv]  
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su  
*Journal of the Royal Statistical Society Series B: Statistical Methodology*, 88 (2): 491–515, Apr. 2026  
Received the 2025 IMS New Researcher Travel Award  
Selected for presentation at JRSSb Editors Invited Session at the RSS Conference 2026
- [J3] **Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts** [arxiv]  
Xiang Li, Garrett G. Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su  
Minor revision at *Journal of the American Statistical Association*.
- [J4] **Debiasing Watermarks for Large Language Models via Maximal Coupling** [pdf] [arxiv]  
Yangxinyu Xie, Xiang Li, Tanwi Mallick, Weijie J. Su, Ruixun Zhang  
*Journal of the American Statistical Association*, 1–21, 2025
- [J5] **Robust Spectral Watermark for Synthetic Tabular Data** [pdf]  
Yizhou Zhao, Xiang Li, Peter Song, Qi Long, Weijie J. Su

- [J6] **Convergence and Inference of Stream SGD, with Applications to Queueing Systems and Inventory Control** [\[pdf\]](#)  
Xiang Li\*, Jiadong Liang\*, Xinyun Chen, Zhihua Zhang  
*Operation Research*, Jan. 2026
- [J7] **Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation** [\[pdf\]](#) [\[arxiv\]](#)  
Yuze Han, Xiang Li, Jiadong Liang, Zhihua Zhang  
*Mathematics of Operations Research*, published online Aug. 2026
- [J8] **Finite-Time Decoupled Convergence in Nonlinear Two-Time-Scale Stochastic Approximation** [\[pdf\]](#) [\[arxiv\]](#)  
Yuze Han\*\*, Xiang Li\*\*, Zhihua Zhang\*\*  
*Journal of Machine Learning Research*, 27 (112): 1–69, 2026.
- [J9] **A Random Projection Approach to Personalized Federated Learning: Enhancing Communication Efficiency, Robustness, and Fairness** [\[pdf\]](#)  
Yuze Han\*\*, Xiang Li\*\*, Shiyun Lin\*\*, Zhihua Zhang\*\*  
*Journal of Machine Learning Research*, 25 (380): 1-88, Dec. 2024
- [J10] **Variance-Aware Decision Making with Linear Function Approximation under Heavy-Tailed Rewards** [\[pdf\]](#) [\[arxiv\]](#)  
Xiang Li, Qiang Sun  
*Transactions on Machine Learning Research*, Apr. 2024  
Invited to present at ICLR 2025; ranked among the Top 100 papers accepted by TMLR that year.
- [J11] **FedPower: Privacy-Preserving Distributed Eigenspace Estimation** [\[pdf\]](#) [\[arxiv\]](#)  
Xiao Guo\*, Xiang Li\*, Xiangyu Chang, Shusen Wang, Zhihua Zhang  
*Machine Learning*, 113 (11): 8427-8458, Sep. 2024
- [J12] **Discussion of “Balanced and Robust Randomized Treatment Assignments: The Finite Selection Model for the Health Insurance Experiment and Beyond”** [\[pdf\]](#)  
Xiang Li, Yangxinyu Xie, Weng-Kee Wong  
*Journal of the Royal Statistical Society Series A: Statistics in Society*, 2026 (Discussion article)

### Conference papers

- [C1] **On the Convergence of FedAvg on Non-IID Data** [\[pdf\]](#)  
Xiang Li\*, Kaixuan Huang\*, Wenhao Yang\*, Shusen Wang, Zhihua Zhang  
*International Conference on Learning Representations (ICLR) 2020* (Oral presentation, Google citation 3300+)
- [C2] **Statistical Estimation and Online Inference via Local SGD** [\[pdf\]](#)  
Xiang Li, Jiadong Liang, Xiangyu Chang, Zhihua Zhang  
*Conference on Learning Theory (COLT) 2022*
- [C3] **Improving the Trade-off Between Watermark Strength and Speculative Sampling Efficiency for Language Models** [\[pdf\]](#)  
Weiqing He\*, Xiang Li\*, Li Shen, Weijie J. Su, Qi Long  
*International Conference on Learning Representations (ICLR) 2026*
- [C4] **Selective Disclosure Watermarking for Large Language Models** [\[arxiv\]](#) [\[pdf\]](#)  
Xuyang Chen, Xiang Li, Yangxinyu Xie, Qi Long  
*International Conference on Machine Learning (ICML) 2026*

- [C5] **On the Empirical Power of Goodness-of-Fit Tests in Watermark Detection** [\[arxiv\]](#)  
Weiqing He\* and **Xiang Li\***, Tianqi Shang, Li Shen, Weijie J. Su, Qi Long  
*Neural Information Processing Systems (NeurIPS) 2025 (Spotlight)*
- [C6] **Mitigating the Privacy–Utility Trade-off in Decentralized Federated Learning via  $f$ -Differential Privacy** [\[arxiv\]](#)  
**Xiang Li\***, Buxin Su\*, Chendi Wang\*, Qi Long, Weijie J. Su  
*Neural Information Processing Systems (NeurIPS) 2025 (Spotlight)*
- [C7] **Steer-to-Detect: Probing Hidden Representations for Detection of LLM-Generated Text** [\[arxiv\]](#) [\[pdf\]](#)  
Luxu Liang, **Xiang Li**  
*Neural Information Processing Systems (NeurIPS) 2026*
- [C8] **Conformal Prediction with Paraphrase-Aware Scoring for LLM Uncertainty Quantification** [\[pdf\]](#)  
Jiayi Xin, Evan Qiang, Zihan Zhu, **Xiang Li**, Weijie J. Su, Qi Long  
*Neural Information Processing Systems (NeurIPS) 2026*
- [C9] **UCS: Estimating Unseen Coverage for Improved In-Context Learning** [\[pdf\]](#)  
Jiayi Xin, **Xiang Li**, Evan Qiang, Weiqing He, Tianqi Shang, Weijie J. Su, Qi Long  
*Findings of the Association for Computational Linguistics (ACL) 2026*
- [C10] **Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards** [\[pdf\]](#)  
Qingyuan Yu, Euijin Baek, **Xiang Li**, Qiang Sun  
*Conference on Uncertainty in Artificial Intelligence (UAI) 2025*
- [C11] **A Statistical Analysis of Polyak-Ruppert-Averaged Q-Learning** [\[pdf\]](#)  
**Xiang Li**, Wenhao Yang, Jiadong Liang, Zhihua Zhang, Michael I. Jordan  
*International Conference on Artificial Intelligence and Statistics (AISTATS) 2023*
- [C12] **Statistical Analysis of Karcher Means for Random Restricted PSD Matrices** [\[pdf\]](#)  
Hengchao Chen, **Xiang Li**, Qiang Sun  
*International Conference on Artificial Intelligence and Statistics (AISTATS) 2023*
- [C13] **Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective** [\[pdf\]](#)  
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang  
*Neural Information Processing Systems (NeurIPS) 2022 (Spotlight)*
- [C14] **Personalized Federated Learning towards Communication Efficiency, Robustness and Fairness** [\[pdf\]](#)  
Shiyun Lin\*, Yuze Han\*, **Xiang Li**, Zhihua Zhang  
*Neural Information Processing Systems (NeurIPS) 2022*
- [C15] **Communication-Efficient Distributed SVD via Local Power Iterations** [\[pdf\]](#)  
**Xiang Li**, Shusen Wang, Kun Chen, Zhihua Zhang  
*International Conference on Machine Learning (ICML) 2021*
- [C16] **Finding Near Optimal Policies via Reducive Regularization in Markov Decision Processes** [\[pdf\]](#)  
Wenhao Yang, **Xiang Li**, Guangzeng Xie, Zhihua Zhang  
*Workshop on Reinforcement Learning Theory, ICML 2021*
- [C17] **Do Subsampled Newton Methods Work for High-Dimensional Data?** [\[pdf\]](#)  
**Xiang Li**, Shusen Wang, Zhihua Zhang  
*AAAI Conference on Artificial Intelligence (AAAI) 2020*

- [C18] **A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning** [\[pdf\]](#)  
Wenhao Yang\*, **Xiang Li\***, Zhihua Zhang  
*Neural Information Processing Systems (NeurIPS) 2019*

### Preprints and submitted manuscripts

- [S1] **Weak Convergence Rates for Partial-Sum Processes of Nonlinear Stochastic Approximation** [\[arxiv\]](#)  
**Xiang Li**, Jiadong Liang, Zhihua Zhang
- [S2] **Optimal Watermark Localization in Mixed-Source Large Language Model Texts** [\[arxiv\]](#)  
Jose H. Blanchet\*\*, T. Tony Cai\*\*, **Xiang Li\*\***, Hao Liu\*\*, Qi Long\*\*, Weijie J. Su\*\*
- [S3] **Optimal Detection for Language Watermarks with Pseudorandom Collision** [\[arxiv\]](#)  
T. Tony Cai\*\*, **Xiang Li\*\***, Qi Long\*\*, Weijie J. Su\*\*, Garrett G. Wen\*\*
- [S4] **Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?** [\[arxiv\]](#)  
**Xiang Li**, Jiayi Xin, Qi Long, Weijie J. Su
- [S5] **Privacy-Preserving Community Detection for Locally Distributed Multiple Networks** [\[arxiv\]](#)  
Xiao Guo, **Xiang Li**, Xiangyu Chang, Shujie Ma
- [S6] **Uncertainty Quantification of Data Shapley via Statistical Inference** [\[arxiv\]](#)  
Mengmeng Wu, Zhihong Liu, **Xiang Li**, Ruoxi Jia, Xiangyu Chang

### Technical reports

- [T1] **Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data** [\[arxiv\]](#) [\[code\]](#)  
**Xiang Li**, Jiadong Liang, Zhihua Zhang
- [T2] **Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: A Jump Diffusion Approach** [\[arxiv\]](#)  
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
- [T3] **Communication Efficient Decentralized Training with Multiple Local Updates** [\[arxiv\]](#)  
**Xiang Li**, Wenhao Yang, Shusen Wang, Zhihua Zhang

### GRANTS & FUNDING

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- Contributor to NIH RO1 Grant Proposal, co-written with Prof. Weijie J. Su & Prof. Qi Long (2024)
  - Assisted in proposal writing, structuring key research points on watermarking.

### PRESENTATIONS

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- *Optimal Detection for Language Watermarks with Pseudorandom Collision*
  - Mar. 2026, ENAR Spring Meeting, Indianapolis.
  - Jun. 2026, ICSA Applied Statistics Symposium, Arlington.
- *Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?*
  - Sep. 2025, AI Research Mixer 2025, UPenn.
- *Robust Watermark Detection and Efficient Proportion Estimation for Human-Edited Watermarked Text*
  - Jun. 2025, ICSA Applied Statistics Symposium, University of Connecticut.
  - Aug. 2025, Joint Statistical Meetings, Nashville, Tennessee.

- *Optimal Robust Detection for Gumbel-Max Watermarks Under Contamination*
  - Jul. 2024, IMS-China International Conference on Statistics and Probability, Yinchuan.
  - Sep. 2024, Penn conference on Big Data in Biomedical & Population Health Sciences.
  - Sep. 2024, Warren & ASSET Center Research Mixer, UPenn.
- *A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules*
  - May 2024, the NUS IMS workshop on statistical machine learning for high dimensional data, Singapore.
  - Jul. 2024, the 2nd joint conference on statistics and data science in China (JCSDS), Kunming.
  - Nov. 2024, the conference on statistical learning and data science (SLDS), Newport Beach, California.
- *Complete Asymptotic Analysis for Projected Stochastic Approximation and Debiased Variants*
  - Sep. 2023, the Allerton conference at the University of Illinois, Allerton Park & Retreat Center.
- *Polyak-Ruppert-Averaged Q-Learning is Statistically Efficient*
  - Jul. 2022, the RL workshop in Shanghai University of Finance and Economics.
- *Statistical Estimation and Inference via Local SGD in Federated Learning*
  - Nov. 2021, the 14-th China-R conference.
- *A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning*
  - Jun. 2019, the machine learning workshop at Peking University.

## TEACHING EXPERIENCES

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### Instructor

- Short course (*Statistical Inference in Large Language Models*) at the 2025 ICSA Applied Statistics Symposium [\[slides\]](#) Summer 2025

### Invited guest lecturer

- STAT 9917: *Statistical Topics in Large Language Models*, UPenn Spring 2025

### Teaching assistant

- *Linear Algebra*, Peking University Spring 2021
- *Reinforcement Learning: Theory and Algorithms*, Peking University Fall 2019

## ACADEMIC SERVICE

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### Area Chair

- ICLR 2027

### Conference review

- NeurIPS, ICML, ICLR, UAI, AISTATS, IJCAI.

### Journal review

- Journal of Machine Learning Research
- Transactions on Machine Learning Research
- Journal of the American Statistical Association
- Information and Inference: A Journal of the IMA
- Annals of Applied Probability
- Operational Research
- IEEE Transactions on Automatic Control
- IEEE Open Journal of Signal Processing
- IEEE Journal on Selected Areas in Communications
- Transactions on Parallel and Distributed Systems
- World Wide Web

### Event organization

- Aug. 2024, Session Chair for “Theoretical and Algorithmic Developments in Federated Learning” at the Modeling and Optimization: Theory and Applications (MOPTA) conference.